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Evidence-Maintained Construction of Geometry Diagram Representations

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Abstract

Reasoners that operate with perceptual data as inputs must construct formal representations that preserve the dependencies and competing hypotheses needed for backtracking. This paper presents the Evidence Maintenance System (EMS), which incrementally constructs formal entities from perceptual inputs while maintaining degrees of belief, source dependencies, and alternative hypotheses. We test EMS on plane-geometry diagrams using all 601 diagrams in the Geometry3K test split and show that it improves representation quality while preserving provenance.

1. Introduction

Automated geometry reasoning has been an AI testbed for decades (Gelernter, 1963; Wu, 1986; Chou et al., 2000). More recent work broadens the task: rather than prove already-formalized theorems, systems must solve textbook problems whose givens and goals are embedded in natural language and diagrams (Seo et al., 2015; Lu et al., 2021). In this setting, applying a theorem is only the last step: a system must first identify the relevant objects, relations, quantities, and goal.

Once a problem is formalized, the reasoning stage can be strong. AlphaGeometry2 couples a neural language model with a symbolic engine and achieves performance above the average human gold medalist on recent International Mathematical Olympiad (IMO) geometry problems (Chervonyi et al., 2025). Recent work identifies unresolved ambiguity in formalized problem inputs as a factor that limits solving (Zhao et al., 2025). The strength of the reasoning stage is exactly what makes the formal input decisive: a reasoner can be sound relative to its premises, while those premises can misrepresent the original problem.

In benchmarks such as Geometry3K, a problem consists of a natural-language question, a geometry diagram, and a set of candidate answers (Lu et al., 2021). For diagram-based problems, the formal input must be constructed from the image. The question and the diagram are different modalities: language parsing recovers one, visual and optical-character-recognition (OCR) processing the other, and their outputs must be aligned to formalize the problem.

In the representative case of Figure 1, the question names the target $m\angle RMN$ and gives that MNPR is a parallelogram, while the diagram shows that Q lies on diagonals RN and MP, that $m\angle QNM$=33°, and that $m\angle MRQ$=38°. These observations must become machine-readable objects, relations, and values for a symbolic reasoner.

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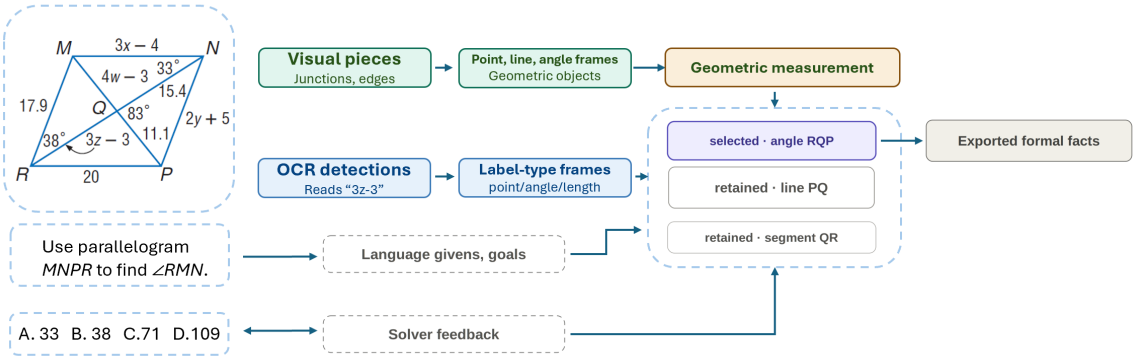


Figure 1. EMS construction for Geometry3K problem 2476. For $3z - 3$, multiple candidate hosts and a geometric measurement are maintained. One host is selected for export while alternatives remain available. Dashed language and solver-feedback paths are future work.

The evidence for this construction is often incomplete or in conflict. Visual marks may be fragmented, OCR engines may disagree on the same token, and a printed measurement may plausibly attach to more than one host. Existing systems such as Inter-GPS (Lu et al., 2021), which is used later as our supervised reference baseline, and PGDPNet (Zhang et al., 2022) address coupled subproblems of primitive extraction, relation recovery, text and symbol recognition, and diagram grounding. Each selected reading and binding becomes a premise of the problem the solver receives.

A natural response is to make perception more accurate. That reduces errors, but it does not preserve the alternatives a later stage may need to revisit a mistaken choice. Existing geometry-solving pipelines typically pass selected relations forward rather than the competing interpretations that led to them (Lu et al., 2021; Zhang et al., 2022). Learned components may assign confidence scores that do not accurately reflect their error rates, especially when inputs differ from the data on which they were trained (Guo et al., 2017; Ovadia et al., 2019). When only a selected prediction is passed to later stages, the alternatives and uncertainty behind that choice are lost. Once those alternatives, and the record of how the choice was made, are discarded, later stages have little basis for reconsidering an incorrect interpretation.

This reframes the difficulty as one of cognitive organization beyond perceptual accuracy. Reasoning is rigorous and perception is steadily improving, but the construction between them runs once and keeps only its result. The question is how that construction should itself be organized and maintained over time.

We address this problem through geometry-diagram understanding: constructing a formal description of the points, segments, angles, incidence relations, measurements, and label-host bindings depicted in a raster diagram. A label-host binding assigns a text label to the geometric object it annotates. Plane geometry is a useful testbed because small errors in reading or binding can change the mathematical meaning, while geometric constraints provide an independent way to

check perceptual decisions. This paper focuses on constructing and maintaining diagram-derived facts; integrating language interpretation and solver feedback is future work.

This paper presents the Evidence Maintenance System (EMS), a framework for constructing formal representations from uncertain diagram evidence within CogSketch. Its central idea is to preserve the evidence and alternatives behind each decision rather than keeping only the final result. EMS retains candidate readings and label-host bindings after selecting a current interpretation, together with their sources and supporting evidence. Visual structure, OCR, and geometric measurement all contribute to this maintained state. In the geometry instantiation, retained candidates can also be used for an inspectable substitution when an exported reading fails a domain-specific consistency check.

EMS builds on the tradition of truth and belief maintenance, in which justifications, dependencies, and alternative contexts are explicit objects (Doyle, 1979; de Kleer, 1986; Provan, 1989). Those systems maintained which beliefs hold; EMS maintains the construction of the symbolic representation itself. Each token reading, object type, label-host binding, and measurement is kept with its sources and retained alternatives, so that the representation handed to a reasoner carries its provenance.

This paper makes three contributions:

1. **Construction as maintained evidence.** We represent the construction of symbolic facts from uncertain visual evidence as an evidential state rather than retaining only the final parse. We instantiate this idea in a working geometry-diagram system that preserves source provenance and competing alternatives throughout construction. For label-host binding, this property is verified by white-box analysis across all 601 test diagrams.
2. **Evidence combination drives the improvement.** A perception-controlled factorial study holds candidate sources fixed and varies only how evidence is organized: winner-take-all selection versus EMS evidence combination. The matched-source contrast therefore isolates the effect of evidence combination from differences in perception. EMS improves structure recall, measurement recall, and measurement F1 under both matched source conditions, with all six within-source paired-bootstrap confidence intervals excluding zero.
3. **Inspectable construction and geometry-specific cross-channel checks.** The maintained state provides corpus-wide provenance and retained alternatives; in the geometry instantiation, these records support analysis of recoverable wrong selections, conservative substitution for malformed angle text, and selective verification of external proposals.

The claims made here have deliberately different scopes: the maintenance architecture is proposed as a general design for representation construction, whereas the empirical evidence establishes only its geometry-diagram instantiation; the angle substitution and verification mechanisms are domain-specific, and language evidence, solver feedback, and transfer to other domains remain future work.

Table 1 places EMS alongside its classical ancestor and modern geometry formalization systems by how each organizes alternatives and what output it emphasizes.

Section 2 presents the evidence-maintenance framework. Section 3 describes its instantiation for geometry diagrams. Section 4 evaluates evidence combination under controlled perception sources. Section 5 examines provenance, retained alternatives, substitution, and selective verification. Section 6 relates EMS to prior work, discusses limitations, and outlines future work.

Table 1. Positioning of EMS relative to its classical ancestor and geometry formalization systems.

System	Primary task	Treatment of alternatives or conflict	Published output emphasized
VICTORS	Model-based visual recognition	ATMS interpretations ranked with DS	Object interpretations
Inter-GPS / PGDPNet	Geometry parsing for solving	Select predicted relations	Formal relation set
Pi-GPS	Rectification for solving	Rectifier plus geometric verifier	Corrected formalization
EMS	Diagram representation construction	Maintained typed candidates plus angle-specific post-check	Formal facts and inspectable formation state

2. The Evidence Maintenance Framework

Reading a geometry diagram is a multimodal problem. Evidence comes from OCR, visual structure, and geometric computation, and these channels must be aligned and fused even when they disagree. A bare number such as 33, for example, may appear to be the length of a nearby side when a dropped degree sign actually makes it an angle measure. EMS therefore needs a calculus that can represent incomplete, uncertain, or conflicting observations. We use Dempster–Shafer theory (Shafer, 1976). The architecture requires such a calculus but does not depend on this particular choice; alternative combination rules remain open (Section 6.4).

A standard probability distribution assigns probability to mutually exclusive atomic hypotheses. EMS also needs evidence that supports a set of hypotheses without choosing among them. Dempster–Shafer mass functions provide this: mass can rest on a set of alternatives, and mass on the full set of alternatives represents unresolved ignorance. Belief and plausibility give lower and upper bounds on the support for a hypothesis, while Dempster's rule combines evidence from separate sources. Keeping both bounds lets EMS represent uncertainty that a single score would collapse.

EMS separates selecting a current interpretation from discarding its alternatives. A downstream reasoner still needs definite facts, so EMS designates some candidates as the current output, while keeping the alternatives, their evidence, and their dependencies. During construction, evidence added before final selection can change candidate rankings at the next explicit maintenance step. This follows the belief-maintenance distinction between a currently accepted belief and the justification structure that supports it (Doyle, 1979; de Kleer, 1986).

The maintained state contains five kinds of objects: evidence sources, typed frames of alternatives, evidence records, candidate interpretations, and derived knowledge. Sources identify where evidence comes from, such as a visual extractor, an OCR engine, or a geometric rule. A frame contains the mutually exclusive alternatives for one decision (Shafer, 1976). Evidence records assign support within a frame, while candidate records summarize the resulting belief in individual alternatives. Knowledge records encode derived relations that later stages can use, such as a point lying on a line. Together, these objects connect each formal diagram fact to the observations and decisions behind it.

The right side of Figure 1 traces this dependency structure for token $3z - 3$ of problem 2476. Textual and visual evidence first develop in separate frames and then meet in a label-host frame. A geometric measurement provides additional visual evidence about which object the text token labels,

a cross-channel constraint. The current-output assignment then selects one host per token for formal export, while the remaining candidates and their provenance stay in the maintained state. In this example, the angle host is selected and the correct segment QR remains available as an alternative. Section 2.3 examines this case in detail.

2.1 Maintained evidential state

Here an evidence record is an EMS data structure, not a standard object of Dempster–Shafer theory: its mass assignment follows the standard calculus, whereas its source, provenance, dependency, and lifecycle fields are EMS additions. An evidence record e has the form

$$e = \langle s, \Theta_e, m_e, c_e, \pi_e, D_e, \sigma_e \rangle,$$

where s is the source, Θ_e is the frame in which the evidence is stated, m_e is its mass assignment, and c_e is the observation confidence. The provenance field π_e links the record to its underlying signal or intermediate result, while D_e records the evidence on which a derived record depends, and σ_e records its stage in the construction lifecycle. The geometry-specific substitution policy records changes separately rather than rewriting provenance (Section 2.3). Each source also has a reliability value, keeping observation confidence distinct from source reliability.

For example, an OCR evidence record contains the engine, recognized text, bounding box, and source confidence for one reading. A derived label-host record points to the text token and to the visual or geometric evidence used to propose its host.

Each hypothesis in a frame has a corresponding candidate record. The record stores its current belief and plausibility, the evidence that supports or opposes it, and its current role in the output. A candidate's status indicates whether it is currently selected or retained as an alternative.

EMS groups alternatives into typed frames of discernment. Each decision receives its own frame Θ_f, a finite set of mutually exclusive hypotheses. The possible readings of an OCR region form a text frame. Point name, angle measure, and length measure form the alternatives in a label-type frame. The possible geometric hosts of a token form a label-host frame. Keeping these decisions in separate frames prevents evidence about different kinds of claims from being combined.

Each evidence record carries a basic probability assignment over its frame: mass may rest on a single hypothesis, expressing specific support; on a subset focal element (a set carrying positive mass), expressing evidence that distinguishes a group of possibilities without resolving the alternatives inside it; or on the full frame, as remaining ignorance.

For a hypothesis A, EMS computes the standard lower and upper support bounds

$$Bel(A) = \sum_{B \subseteq A} m(B), \qquad Pl(A) = \sum_{B \cap A \neq \varnothing} m(B).$$

Belief, the lower bound, is the conservative score on which EMS ranks and selects.

Frames appear as construction creates the alternatives that populate them. A label-host frame, for example, cannot form until candidate points, lines, or angles exist. This staged introduction lets coarse evidence stay unresolved until the objects required for a finer decision have been built.

2.2 Belief combination and dependency control

Before combining evidence, EMS discounts each mass assignment by the product of source reliability and observation confidence (Shafer, 1976). Discounting weakens specific support by moving the removed mass to ignorance. It does not transfer that mass to competing hypotheses, so an unreliable source contributes little information. Discounted assignments on the same frame are combined using normalized Dempster's rule. EMS retains the raw pairwise conflict for diagnostics, even though the resulting assignment is normalized over the nonconflicting mass. If conflict is effectively total, EMS returns a vacuous assignment (all mass on the frame) rather than performing an unstable normalization. Multiple records on the same frame are combined by repeated application of this rule.

Dempster combination is intended for distinct bodies of evidence (Shafer, 1976), so EMS controls which records may combine. Derived records retain their dependencies and may also belong to a dependency group. Within a frame, EMS uses at most one record from each group. When duplicate derivations occur, it keeps the highest-confidence representative. Support relations that can feed back into one another are placed in the same group, so that the same observation is not counted twice through a derivation cycle. These controls handle obvious dependencies, but they do not fully capture dependence among measurements derived from the same image (Section 6.4). This separation also keeps source evidence distinct from later interpretations: fused token readings, type hypotheses, and label-host hypotheses are created in later frames, each linked to its sources.

2.3 Staged construction, maintenance, and selection

Problem 2476 illustrates why auditability is useful even when the selected output is wrong. The printed expression $3z - 3$ denotes the length of segment QR. EMS instead selects an angle host. A selected-output evaluation records this as an error, correctly.

The maintained frame reveals more (Figure 2). Candidate beliefs come from a single-diagram run, and the label-host frame is determined before the post-selection angle check, so its contents are unaffected by that policy. The two highest-belief candidates, lines PQ and PR, have already been claimed by the tokens 11.1 and 20 during the global exclusive assignment. Among the remaining hosts, angle RQP is selected, while the correct segment QR remains a lower-belief alternative.

This case makes the selection stage concrete: OCR and geometric evidence populate the label-host frame, maintenance recomputes its beliefs, and exclusive selection chooses one available host while preserving the remaining candidates.

Construction is staged. Evidence combines only within a frame, while derived records carry support to later frames and preserve the dependency path. The current implementation separates evidence insertion, maintenance, and selection (Algorithm 1). Adding evidence does not recompute belief. Instead, maintenance is invoked explicitly at fixed pipeline boundaries and recomputes every frame from active, dependency-controlled evidence. This batch design matches the staged pipeline, where evidence is introduced in groups at construction boundaries; incremental dependency propagation is compatible with the framework but is not implemented here. Selection likewise occurs at fixed construction points rather than on demand.

Algorithm 1. Current EMS control flow.

CONSTRUCT(input)

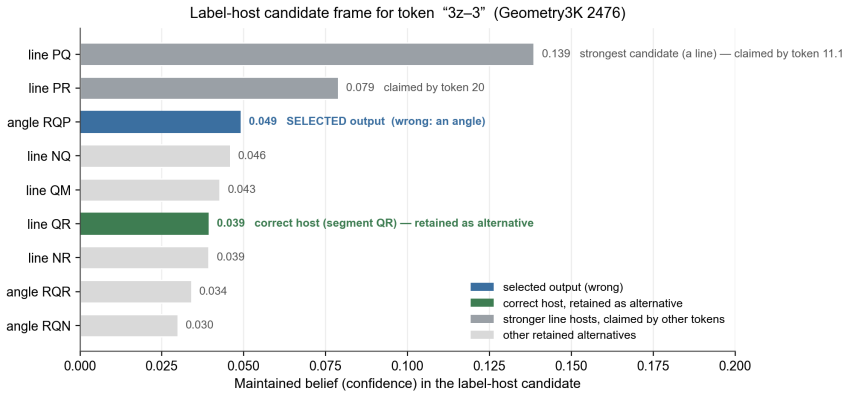


Figure 2. Maintained host candidates for $3z - 3$. Length candidates rank highest but are claimed by other tokens under global exclusivity, leaving an angle selected while the correct segment QR remains retained.

for each scheduled construction stage:
    ADD-EVIDENCE(state, evidence produced by the stage)
    at fixed pipeline boundaries: MAINTAIN(state)
MAINTAIN(state)
SELECT-CURRENT-OUTPUTS(state)
while line extension produces new candidates:
    MAINTAIN(state)
    SELECT-CURRENT-OUTPUTS(state)
ANGLE-POSTCHECK(state) // geometry-specific; no maintain/reselect
run coverage/rendering diagnostics
EXPORT(state)
ADD-EVIDENCE(state, e)
store e with its frame, dependencies, and provenance
ensure candidates for hypotheses supported by e
MAINTAIN(state)
for each frame f:
    choose one active record per dependency group
    combine discounted evidence
    update candidate belief and plausibility
SELECT-CURRENT-OUTPUTS(state)
commit candidates above threshold
greedily enforce one-to-one assignment for label-host bindings
mark chosen bindings selected-final
retain competing bindings as alternatives

Evidence added before final selection can change candidate scores and the selected binding. Geometric angle measurements are inserted before reconstruction for this purpose. After final selection, the core maintenance loop is not re-entered: the geometry-specific angle check of Section 3.5 reads the retained state without maintenance or reselection. Non-selected candidates and their provenance remain available for inspection.

3. Instantiating EMS for Geometry Diagrams

This section describes how EMS constructs a formal geometric representation from visual pieces, while maintaining the evidence behind each decision. It specializes the framework of Section 2 to raster plane-geometry diagrams, fixing what counts as a point, line, token, and measurement. The resulting diagram-derived facts can later be combined with language by a symbolic reasoner. Each stage maintains its own evidence: visual pieces, OCR readings, geometric measurements, and label-host bindings, with their sources and retained alternatives.

The implementation is written in Common Lisp and runs within CogSketch (Forbus et al., 2011). Its input adapters, evidential state, geometric construction rules, selection procedure, and exporter for formal facts are separate modules. Geometry3K logic forms are used only by the evaluation scripts, not by the diagram pipeline while it constructs a representation.

3.1 From raster input to visual evidence

Rather than converting the image directly into geometry, EMS first builds an intermediate layer of visual pieces between raster pixels and recognized objects, which can be recomputed as interpretation proceeds. Following the Hybrid Primal Sketch, where low-level vision yields sketch-like entities for a higher-level interpreter (Forbus et al., 2024), EMS treats these pieces as evidence rather than settled objects. A preprocessing stage erases detected text, traces the remaining ink into SVG vector contours, and decomposes them into edges, segments, and junctions (Appendix B). At this stage, long contours may support lines, curved pieces may support circles, and intersections or endpoints may support points.

Each imported piece becomes a visual evidence record linked to its originating SVG object and carrying a source identifier, confidence, and payload. Junctions support point hypotheses, curved edges support circle-fit hypotheses, and pairs of supported points support line hypotheses. For lines, the evidence includes how well the connecting segment is covered by observed ink. The diagram thus enters the system as evidence within typed frames, each recomputable as more objects appear.

EMS constructs objects in an order that respects their dependencies and the strength of available evidence. Points and lines supported by strong evidence are built early, so that later decisions, such as measurement binding, have candidate hosts to refer to. Circle hypotheses are considered before line hypotheses, because their arc pieces would otherwise support many spurious lines, and a maintained circle protects its supporting edges from the line proposer. Lines are then proposed between point candidates. Conservative, experimenter-set coverage gates govern circle and line admission, preferring missed faint structure to spurious structure (Appendix B).

3.2 OCR evidence and label types

The OCR adapter separates locating text from reading it. A full-image detection pass first finds the image regions that contain text, fixing where each text cluster sits. Several OCR engines then read each located region, and their outputs can confirm or add alternative readings for the detected region. Released Inter-GPS perception records enter through this same secondary-source interface in one experimental condition, and they do not replace the EMS pipeline. The engines and clustering rules are listed in Appendix B.

Raw engine outputs are kept as records of what each engine read in each region. A fused token combines these outputs while retaining links to the raw records, so selecting one reading does not discard the alternatives (Appendix B). The fused readings then provide evidence for a label-type frame. A single uppercase letter can support a point-name hypothesis, while a degree-marked number supports an angle-measure hypothesis. When a numeric or algebraic expression does not distinguish between an angle and a length, its mass can be assigned to the subset containing both hypotheses. These conventions enter as rule-based evidence with provenance and influence type selection without becoming irreversible declarations.

3.3 Constructing lines, angles, and subsegments

Line construction combines point support with visual coverage. The system records point-on-line relations when an interior point lies on a proposed line. These relations are used after the first reconstruction pass: if a selected line contains a selected interior point, EMS introduces its two subsegments as additional line candidates. A measurement printed beside half a diagonal can then bind to the subsegment it actually annotates rather than the full diagonal. Existing evidence determines whether that subsegment is selected.

Angles are generated from line pairs sharing an endpoint. An angle record identifies its vertex and legs and caches the angle value from their coordinates. This geometric value is not treated as the printed measurement. Instead, it provides a separate constraint that can favor one label-host alternative or expose disagreement with an OCR reading.

After reconstruction, the pipeline compares the selected geometry with the visual evidence and measures how much line-like ink remains unexplained. When coverage is below target, unexplained pieces generate residual evidence. Line proposals are then reconsidered under gradually relaxed coverage gates. The relaxation schedule and stopping criteria are in Appendix B.

3.4 Label-host binding

For each token, EMS constructs a label-host frame containing nearby point, line, subsegment, and angle hosts of compatible types. Position, inferred label type, host type, and geometric compatibility contribute explicit evidence to that frame. For an angle-bearing token, for example, the difference between its numeric OCR value and the angle value computed for each host contributes strong, weak, or vacuous support. A large discrepancy does not falsify a host; instead, it adds support to a geometrically matching alternative that can overtake the current candidate.

The displayed interpretation and the formal output require an exclusive assignment. EMS obtains a one-to-one output assignment by greedily selecting the highest-belief available token–host pair, so that a token takes at most one host and a host at most one token. One-to-one assignment matches the output convention of Geometry3K annotations; greedy selection is an implementation choice rather than part of the framework. This global constraint can make the selected output differ from the locally highest-scoring candidate. Non-selected candidates remain in the state with alternative status, making the global assignment's effect inspectable.

3.5 Post-selection angle check and formal export

Before formal export, the geometry instantiation performs one angle-specific consistency check on the selected output. This check is not part of the domain-general maintenance process: it consults the maintained OCR candidate pool but does not add evidence or rerun maintenance or selection. Because diagrams may not be drawn to scale, geometry serves only as a consistency constraint. The system records a conflict when the angle text cannot be parsed or when the OCR reading differs from the geometric measure by more than 45 degrees. Substitution is allowed only when the original literal is malformed or out of range and a retained degree-marked OCR candidate is repeatedly observed, dominant, and within 8 degrees of the geometric measure. A valid in-range literal is never overwritten. Each decision records the original reading, any replacement, the geometric measure, the alternatives considered, and the source token.

EMS then translates the selected geometry and label-host bindings into predicate-calculus assertions for points, segments, incidence relations, angles, measurements, and links to source tokens and binding confidences. The evidential state remains intact for inspection.

4. Evaluation: Does Evidence Combination Improve Construction?

This section evaluates whether evidence combination improves representation construction when candidate perception is held constant. We use a perception-controlled design (Section 4.2) and ask two questions: whether evidence combination improves construction under fixed candidate sources, and whether an enriched candidate source can substitute for it.

4.1 Data and metrics

We evaluate all 601 diagrams in the standard Geometry3K test split, problem identifiers 2401 through 3001 (Lu et al., 2021). Every condition uses the same images and evaluation targets. The diagram logic forms are consulted only after the formal facts are exported.

We report two micro-aggregated axes, pooling counts across all diagrams. Structure recall combines named points, segment connectivity, and point-on-line incidence, so it depends on both geometric construction and sufficient naming rather than detector performance alone. Named-point and connectivity precision are reported in Table 4. Incidence is evaluated only against annotated ground-truth relations: the evaluator checks whether each annotated incidence is supported by EMS's collinearity closure, but does not enumerate a corresponding prediction-side incidence set. Therefore, no unified structure precision is defined.

Measurement recall and precision count an emitted measurement as correct only when both its normalized value and its length or angle host are correct. Measurement F1 is the harmonic mean of the corpus-level precision and recall. Relational equalities between two unknowns are excluded from this axis because the system extracts printed values rather than solving relations.

The coverage and admission thresholds in Appendix B were fixed during development on training-split diagrams before any test-split runs. The 45-degree conflict threshold was based on documented OCR failure modes rather than test outcomes. The conservative substitution policy of Section 5.3,

Table 2. Full-test-split 2×2 evaluation. Bold marks EMS gains over WTA within each source condition. Runs predate the post-selection substitution policy; the matched replay of Section 5.3 leaves results unchanged within rounding.

	Perception sources	Rule	Structure R	Measurement R	Measurement P	Measurement F1
A	Our stack	WTA	0.183	0.110	0.645	0.189
B	Our stack	**EMS**	**0.288**	**0.149**	0.517	**0.231**
C	Supplemented	WTA	0.184	0.099	0.597	0.170
D	Supplemented	**EMS**	**0.297**	**0.152**	0.500	**0.233**

including its 8-degree tolerance, dominance gate, and support gate, was designed after inspecting conflict events and is therefore excluded from this section's primary performance claims.

4.2 Perception-controlled design

Perception quality and evidence combination cannot be separated in a simple end-to-end comparison, so we control them experimentally. The design holds candidate generation, admission, normalization, and assignment fixed, and varies only the combination rule. This isolates the effect of evidence combination under matched candidate sources. The effects of retaining provenance and alternatives are evaluated separately in Section 5. The experiment crosses two source conditions with two combination conditions.

The source conditions are our stack (the CogSketch, PaddleOCR, secondary OCR, and geometry pipeline of Section 3, run zero-shot with no task-specific training) and a supplemented set that adds cached Inter-GPS RetinaNet and Mathpix records as secondary evidence. The two combination conditions are winner-take-all (WTA) and EMS. WTA keeps the strongest individual non-ignorance focal element without accumulating corroborating evidence. EMS applies normalized Dempster combination over the same discounted evidence while retaining the candidate distribution.

Both arms use the same greedy one-to-one assignment over belief-ranked pairs (Section 3.4). Under WTA, belief collapses to the winning element, reducing the ranking information available to the assignment procedure. This loss of ranking information is part of the WTA ablation. Since the assignment procedure is otherwise identical, the contrast reflects the evidence-combination rule rather than differences in assignment. These choices produce cells A through D in Table 2.

For uncertainty estimates, each within-source contrast uses a paired bootstrap over the 601 test diagrams: the diagrams are resampled with replacement 10,000 times, and in each resample both conditions are evaluated on the same sampled diagrams, yielding 10,000 matched differences. All six within-source contrasts (three metrics for B-A and D-C) have 95% intervals that exclude zero, ranging from +0.038 to +0.113; Table 3 reports the full intervals.

4.3 Results of the controlled comparison

Evidence combination improves all three evaluated metrics under both source conditions (Table 2). With our stack, measurement F1 rises from 0.189 to 0.231. With the supplemented source set, it rises from 0.170 to 0.233. Structure recall shows larger gains under both source conditions.

Table 3. Within-source effects of replacing WTA with EMS evidence combination. All six intervals exclude zero.

Contrast	Metric	Difference	95% paired-bootstrap interval
B - A	Structure recall	+0.1049	[+0.0898, +0.1211]
B - A	Measurement recall	+0.0381	[+0.0223, +0.0543]
B - A	Measurement F1	+0.0422	[+0.0191, +0.0658]
D - C	Structure recall	+0.1129	[+0.0962, +0.1308]
D - C	Measurement recall	+0.0531	[+0.0350, +0.0712]
D - C	Measurement F1	+0.0635	[+0.0372, +0.0894]

The supplemented condition clarifies attribution. Under WTA, adding the supervised records leaves structure recall essentially unchanged and lowers measurement recall from 0.110 to 0.099. Under EMS, the same supplement leaves measurement recall (+0.0034, 95% CI [-0.0055, +0.0123]) and measurement F1 (+0.0023, [-0.0108, +0.0154]) essentially unchanged, and produces only a marginal increase in structure recall (+0.0088, [+0.0005, +0.0193]). The supplement is therefore a control, not a performance booster. The robust effect is the matched-source contrast: the enriched candidate set yields substantially better measurement output under EMS than under WTA, so source enrichment alone does not explain the gain.

In absolute counts, EMS raises correctly matched measurements from 162 to 218 (A→B) and 145 to 223 (C→D). WTA's higher precision follows from its conservative output, emitting far fewer measurements per cell. EMS recovers these correct facts at the cost of more false positives, and the F1 improvement shows the gain is not indiscriminate emission.

Structure recall moves the most because the metric depends on both reconstructed geometry and sufficient OCR-supported naming, both of which evidence combination can affect. Unlike the measurement axis, where EMS trades precision for recall, the structure axis gains recall at essentially constant precision: named-point precision moves from 0.874 to 0.867 while recall rises from 0.330 to 0.476 (Table 4 reports all four cells). The measurement axis is nevertheless the more direct test because it requires both the correct value and the correct host, and EMS improves it under both source conditions.

Table 4. Structure-axis precision and recall by cell.

Cell	Points P	Points R	Connectivity P	Connectivity R
A (our stack, WTA)	0.874	0.330	0.916	0.146
B (our stack, EMS)	0.867	0.476	0.922	0.247
C (supplemented, WTA)	0.877	0.334	0.923	0.146
D (supplemented, EMS)	0.866	0.476	0.918	0.256

Connectivity precision is computed as one minus the fraction of emitted segments counted as extra. The evaluator credits ground-truth segments covered by collinear closures, so its matched count can exceed the number of emitted segments; a small number of emitted segments that span an

interior point are therefore counted as extra although the recall side credits them, making connectivity precision a conservative lower bound. Incidence is scored only from the ground-truth side, so incidence precision is unavailable.

Table 5 provides context rather than a second causal test. Inter-GPS is a supervised geometry problem-solving system whose parser uses a trained detector and Mathpix OCR (Lu et al., 2021). We also evaluate three open-weight VLMs—Qwen2.5-VL:7B (Bai et al., 2025), LLaVA:7B (Liu et al., 2023), and MiniCPM-V (Yao et al., 2024)—as image-only, zero-shot baselines prompted to identify points, segments, and bound measurements. Because their perception methods and training differ from EMS, these results provide reference points rather than isolating the effect of the maintenance framework.

Table 5. Reference results on the Geometry3K test split. Bold marks recall for the supervised parser and the strongest VLM, and precision for the EMS cells. "Not exposed" refers to the evaluated selected-output interface, not to whatever internal diagnostics each system may keep.

System	Diagrams parsed	Measurement R	Measurement P	Maintained state exposed in evaluated output
Inter-GPS released parser	601	**0.426**	0.413	Not exposed
Qwen2.5-VL:7B	579	**0.163**	0.166	Not exposed
LLaVA:7B	562	0.003	0.002	Not exposed
MiniCPM-V	520	0.019	0.046	Not exposed
EMS cell B	601	0.149	**0.517**	Yes
EMS cell D	601	0.152	**0.500**	Yes

Inter-GPS achieves higher measurement recall and F1, as well as stronger reported point and segment parsing results; it is a supervised parser, whereas EMS is zero-shot. For this paper, the more important distinction is the state each system exposes. EMS achieves higher measurement precision in this evaluation and, more importantly, preserves the competing candidates and sources behind selected label-host bindings. Among the VLMs, Qwen2.5-VL slightly exceeds EMS in measurement recall but at roughly one-third of its precision, while LLaVA and MiniCPM-V produce few correct bound measurements. These results motivate the selective verification experiment in Section 5.4.

5. Auditability, Alternatives, and Cross-Channel Checks

This section examines what the maintained state contains and what behavior it supports. We examine four properties in turn: corpus-wide provenance and alternatives (Section 5.1), recoverability of wrong selections (Section 5.2), conflict detection and conservative substitution (Section 5.3), and a selective gate over external proposals (Section 5.4).

Table 6. Corpus-wide properties of the maintained state. Bold marks the principal audit and replay outcomes discussed in the text.

Property	Full-test-split result
Cell B white-box audit (main experiment build)	
Selected label-host bindings	2,996
Bindings traceable to a source OCR token	**2,996 / 2,996 (100%)**
Retained alternative OCR readings	13,388
Mean retained readings per selected binding	**4.47**
Mean name-independent visual coverage	0.723
Diagrams with coverage at least 0.70	58.7%
Matched-perception policy replay (perception rerun; Section 5.3)	
Conflict-marked decisions	**105 in 72 / 601 diagrams (12.0%)**
Automatic changes	**11**
of which value-changing substitutions	3
of which same-value canonicalizations	8

5.1 Corpus-wide maintained state

Table 6 reports two analyses over the full 601-diagram test split, both run with production defaults and without ground-truth access: a white-box audit of cell B (our stack under EMS) and the matched-perception policy replay of Section 5.3.

This analysis is possible because EMS preserves provenance and alternatives rather than exposing only the selected output. The two blocks come from separate runs, and the replay reran perception, so their absolute counts should not be compared directly. Every selected label-host binding can be traced to a fused token and, through it, to one or more raw OCR observations. This traceability is reported here for bindings; other exported facts retain their own support records. The mean of 4.47 retained readings indicates how many alternatives remain available for inspection and substitution, not how many readings are equally plausible.

The visual-coverage diagnostic helps locate a major bottleneck. The system explains 72.3% of line-like visual evidence on average, whereas named structure recall in cell B is 28.8%: many lines are reconstructed but cannot enter a name-dependent formal relation without an OCR label. The gap tracks the dependence on OCR naming and on current marker recognition.

5.2 Recoverability of wrong selections

Across the 314 evaluable exported measurement bindings in a matched-perception run, 221 were correct. Among the 93 wrong-host cases, the correct host remained in the token's frame in 33 (35.5%). When the correct host existed in reconstructed geometry, it reached the relevant frame in 33 of 36 cases (91.7%). Retained correct hosts also ranked highly (median rank 3; 93.9% in the top five). Thus selection can be wrong while the material for a later repair remains represented and traceable. Appendix A gives the protocol, scope caveats, sensitivity analyses, and reading-axis results.

5.3 Conflict detection, conservative substitution, and abstention

Section 3.5 defines the geometry-specific post-selection policy. Cross-channel conflicts occur in 72 of the 601 diagrams (Table 6). In a matched-perception replay, a legacy rule that overwrites whenever geometry disagrees made 97 automatic changes, whereas the conservative policy made only 11 while preserving the aggregate measurement result.

A controlled injection experiment on training-split problem 224 tests both repair and abstention: the retained OCR pool contains the correct 85° reading while the fused winner is perturbed. The policy repairs an out-of-range 185° reading but preserves a clean, range-valid 65° reading despite geometric disagreement. A natural case in problem 2962 shows the same pattern, and all three naturally occurring value-changing repairs in the matched replay were correct in a post hoc audit. Because the policy was designed after inspecting these events, we treat this as evidence about precision and safety rather than an unbiased repair-accuracy estimate. Appendix A reports the complete conditions and diagnostics.

5.4 Selective verification of external proposals

EMS can also use reconstructed angle geometry as a conservative acceptance gate for external proposals. On a 200-problem test subset, the released Inter-GPS parser produced 289 angle proposals, 51 of which were correct (precision 0.176). The geometric gate accepted only 15 proposals, 12 of them correct (precision 0.800), trading recall for a small, geometrically supported, and auditable subset. Table A2 in Appendix A reports the same procedure for proposals from three VLMs.

6. Related Work, Limitations, and Future Directions

6.1 Belief maintenance and visual interpretation

Truth Maintenance Systems made accepted beliefs depend on explicit justifications, the ATMS maintained the environments in which conclusions hold, and Falkenhainer's Belief Maintenance System extended this idea to graded uncertainty (Doyle, 1979; de Kleer, 1986; Falkenhainer, 1988). EMS follows this tradition but applies maintenance to a different problem: constructing a formal representation from uncertain perceptual evidence. In this setting, discarding an alternative can silently change the mathematical problem given to the reasoner.

VICTORS is the closest classical visual ancestor. Provan (1989) embedded Dempster–Shafer belief functions in an ATMS-based recognition system to maintain and rank multiple visual interpretations. EMS instead maintains the construction process that turns heterogeneous perceptual observations into formal premises. Its state includes typed decision frames, derived evidence across frames, dependency-group controls, and selected facts that remain linked to their formation history.

6.2 Perceptual and evidential representations

CogSketch provides qualitative visual processing and relational representations for sketches (Forbus et al., 2011), while the Hybrid Primal Sketch adds computer-vision components that produce sketch-like entities for conceptual interpretation (Forbus et al., 2024). EMS builds directly on this line of work. It follows the same progression from raster input, through sketch-like intermediate pieces, to

conceptual structure, while maintaining uncertainty over the decisions that connect those pieces to formal facts. Unlike the Hybrid Primal Sketch, which commits to an interpretation of each piece, EMS retains competing interpretations and their sources for later inspection and reconsideration.

Belief functions have also been applied directly to image interpretation. Evidential grammars, for example, combine uncertain segment classifications with structural rules to search for coherent scene interpretations (Bordes et al., 2013). EMS differs in both its task and the state it maintains: its goal is to construct a structured diagram representation for later reasoning, while preserving candidate readings and bindings so they remain available and traceable after selection.

6.3 Geometry formalization and correction

Inter-GPS parses Geometry3K text and diagrams into formal logic forms before symbolic theorem application (Lu et al., 2021). PGDPNet uses instance segmentation and graph reasoning to recover diagram primitives and relations (Zhang et al., 2022). GeoX applies formalized vision-language pretraining (Xia et al., 2024), while Geoparsing learns unified formal descriptions for plane and solid geometry (Wang et al., 2026). These systems focus on producing a selected formal description. Learned components may also assign confidence scores that do not accurately reflect their error rates, especially when test inputs differ from their training data (Guo et al., 2017; Ovadia et al., 2019). EMS asks a different question: what information should be preserved while such a description is being constructed? It retains competing readings and their sources after selection, leaving later stages with alternatives that can be inspected and reconsidered. EMS is therefore complementary to discriminative parsers rather than a replacement for them.

Pi-GPS resolves textual ambiguity with a multimodal rectifier and uses a geometric verifier to suppress invalid corrections, producing a corrected input for solving (Zhao et al., 2025). EMS instead focuses on the construction state itself, retaining both selected and nonselected interpretations together with their sources. AlphaGeometry2 shows how powerful symbolic reasoning can be once an appropriate formal statement is available (Chervonyi et al., 2025). EMS addresses part of the preceding problem of constructing that statement.

Table 1 compares these systems in how they organize alternatives and what outputs they expose. Among the systems considered here, EMS is the one that makes the intermediate construction state its primary object of study.

6.4 Limitations

(1) Scope. EMS currently maintains only diagram-derived evidence; integrating language evidence and solver feedback remains future work.

(2) Perception. The zero-shot stack misses some geometric markers and unnamed structure. OCR naming, hard admission gates, and greedy assignment also limit recall.

(3) Uncertainty modeling. Dependency controls prevent obvious double counting but do not fully capture dependence among observations from the same raster image. Alternative combination rules and source calibration remain open.

(4) Post-selection policy and evaluation. Substitutions are recorded rather than fully retractable. The post hoc policy still requires an independent frozen evaluation, and the external verification gate remains narrow.

6.5 Future directions and conclusion

The next step is to add language-derived givens and goals as evidence sources over compatible frames. A statement such as "segment QR has length $3z - 3$" could then act directly on the retained host alternatives described in Section 2.3. A symbolic solver could provide another source of feedback: failure to derive a result or contradiction among premises could trigger targeted reconsideration rather than a complete reparse. These extensions would test whether the maintained representation can persist across perception and reasoning.

EMS should also be tested with stronger open perception sources and in diagram domains beyond plane geometry. When a source exposes top-k readings rather than only its argmax, those candidates can enter the maintained state directly. Selective verification (Section 5.4) also suggests a near-term use as an auditable check on model-proposed facts. Within EMS, hard admission gates should move toward explicit uncertainty, dependence handling should be strengthened, and substitutions should become retractable maintained objects.

Together, these extensions point toward a single maintained state spanning perception, language, and symbolic reasoning. This study establishes a narrower result: geometry-diagram construction can be organized as evidence maintenance, and under matched perception sources, evidence combination improves recall and F1 over winner-take-all selection while preserving provenance and alternatives. In the geometry instantiation, cross-channel checks remain inspectable, and the conservative substitution policy repairs malformed readings without altering the exported representation.

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Appendix A. Conflict Repair, Recoverability, and Selective Verification Details

Table A1. Controlled substitution demonstration on training-split problem 224. The geometry-derived value is $86.92°$; the retained OCR pool contains the original $85°$ reading in all conditions. Under the conservative policy, (c) is a substitution of an out-of-range reading, while (d) is a deliberate abstention: a clean, range-valid literal is preserved, and the decision is retained as an inspectable record rather than overwritten.

Condition	Fused text perturbation	Substitution enabled	Exported angle
(a) Baseline	None	Yes	$85°$
(b) Leading-digit error	$85 \rightarrow 185$	No	$185°$
(c) Leading-digit error	$85 \rightarrow 185$	Yes	$85°$
(d) Range-valid reading	$85 \rightarrow 65$	Yes	$65°$

Condition (b) shows the counterfactual need for the mechanism: with substitution disabled, the erroneous $185°$ reading silently enters the formal representation. In condition (c), because the injected $185°$ is out of range, the geometry-text conflict triggers a search over the retained OCR readings, restores $85°$, and records both readings, the geometric measurement, residuals, and source token; the residual of about 98 degrees is retained as a diagnostic rather than serving as the trigger. Condition (d) shows the limit the policy deliberately respects. The injected 65 is a clean, range-valid reading whose residual against the wedge is about 22 degrees, below the 45-degree gross-conflict threshold; because a diagram need not be drawn to scale, the system preserves the literal. The decision, its residual, and the retained candidate pool are kept as an inspectable decision record, and no conflict is marked. The policy thus repairs a grossly inconsistent reading but abstains on a plausibly valid one, even when geometry would prefer a different value.

Selective verification extends to open VLMs on a separate 112-diagram development corpus drawn from the training split (Table A2). The gate rejects almost all proposed angles, including all 110 incorrect LLaVA proposals. Qwen2.5-VL's accepted precision rises from 0.113 to 0.667, and MiniCPM-V's from 0.015 to 1.000, but only three and one proposals, respectively, survive.

Recoverability protocol (Section 5.2 of the main paper). The statistics are a descriptive audit of one full matched-perception run of the current build. The run reproduces the archived cell-B measurement result closely (221 matched here versus 218 archived) but is separate from both analyses reported in Table 6 of the main paper; its counts should not be mixed with either the archived cell-B audit or the policy-replay counts. Of 1,222 selected measurement bindings, 799 had host endpoints without OCR letters and were never exported; a further 32 fall in diagrams whose ground-truth logic

Table A2. Selective geometric verification of VLM angle proposals on the 112-diagram development corpus.

Source	Naive accepted / correct / P	Verified accepted / correct / P
Qwen2.5-VL:7B	71 / 8 / 0.113	3 / 2 / 0.667
LLaVA:7B	110 / 0 / 0.000	0 / 0 / 0.000
MiniCPM-V	67 / 1 / 0.015	1 / 1 / 1.000

form records no value-carrying measurement at all, leaving no host to identify; 75 had printed values absent from the ground truth; and 2 had degenerate keys, leaving 314 evaluable exported bindings.

A host counts as ground-truth compatible when its value matches the annotated measurement; in 18 cases several ground-truth hosts share a value, and a hit on any of them counts. Restricting to the 296 bindings whose value identifies a unique ground-truth host leaves the result unchanged (29 of 83 in-frame, 34.9%, with 29 of 29 conditioned on constructed geometry).

The 93 wrong-host count is a prediction-side quantity and is not comparable with the ground-truth-side error counts of the evaluator. On the reading axis, among 34 cases with a correct host and a wrong printed value, the correct reading remained in the retained OCR pool in 16 (47.1%, [31.5, 63.3]). Bracketed intervals are 95% Wilson binomial confidence intervals. These are descriptive corpus statistics rather than hypothesis tests.

Appendix B. Implementation Details

This appendix specifies the engineering parameters of the Section 3 diagram pipeline, whose Common Lisp implementation in CogSketch (Forbus et al., 2011) separates input adapters, evidential state, geometric construction rules, selection, and formal-fact export; Geometry3K logic forms are confined to evaluation and unavailable during representation construction.

Raster preprocessing (Section 3.1 of the main paper). An OCR and preprocessing service produces bounding boxes used to erase text, preventing letter and numeral strokes from becoming geometric lines; Potrace then traces the erased image, and CogSketch decomposes its contours into edges, segments, and junctions and computes basic spatial information.

Coverage gates (Section 3.1 of the main paper). Compatible arc fits form a circle candidate only when their union explains at least 70% of a circumference, after which maintained circles above the claim threshold protect their supporting edges from the line proposer; lines proposed between point candidates initially require 70% visual coverage.

OCR sources (Section 3.2 of the main paper). Full-image PaddleOCR detections anchor spatial clusters, which local re-recognition, EasyOCR, and formula-oriented OCR can confirm or enrich; a secondary detection outside every anchor may form a discovery cluster. Voting uses canonicalized text, source confidence, and rules that keep repeated fragments from displacing mathematical expressions.

Residual iteration (Section 3.3 of the main paper). When coverage is below target, line proposals are reconsidered under coverage gates relaxed to a 40% floor; iteration stops at 90% coverage, when a pass adds no new residual evidence, when coverage decreases, or after six passes.

Reproducibility. All 2,404 diagram-condition runs completed without runtime errors; evaluation records, outputs, and the bootstrap script (seed 20260609) are retained in the project archive.